AI 101 for Mechanical Engineers · Module 15

Responsible use and human oversight

Lesson 15.3

What you will learn

> Recognize unsupported AI-generated claims, protect restricted information, document permitted assistance, and define specific human responsibility and authority to intervene before AI-assisted work affects an engineering decision.

Lesson overview

Module 15 · Lesson 3

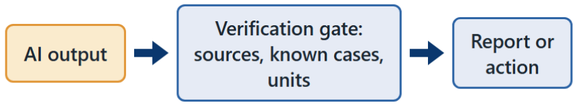
Responsible use and human oversight

AI 101 for Mechanical Engineers

1 Check output integrity

Hallucination (confabulation): plausible but false or unsupported output, such as

- an invented standard citation
- an incorrect material property
- a flawed equation



A fluent answer or a citation label is not evidence.

2 Three meanings of "bias"

Data-coverage bias
Relevant machines, people or operating conditions are missing or under-represented.

Social fairness
Unequal performance or impact across people or groups.

Bias term
The additive parameter b in a neuron.

Different ideas need different analyses. Do not treat evidence about one as evidence about another.

3 Make oversight specific

"Human in the loop" is not a control unless responsibility is specific.

1. Name the reviewer and the decision owner
2. State which outputs need review
3. Define the evidence to check
4. Give a named person authority to pause or stop
5. Keep an audit trail: data, assumptions, units, baselines, decisions

4 Protect information and systems

Do not share restricted personal, assessment or proprietary data without permission and safeguards. Review generated code and connected tools before integration; use source control and logs.

5 Resources and disclosure

AI computation uses physical resources: scale the method to the task. Disclose AI assistance as policy requires and keep a trail separating evidence, generated suggestions and your decisions.

Key takeaway

Verify AI output, protect information, and name who is responsible.
Oversight works only with a named reviewer, defined checks and the authority to stop.

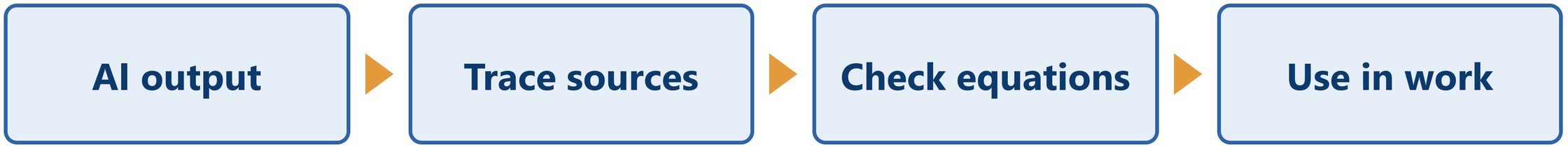
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Fluent output can still be false

- Hallucination or confabulation: plausible but false or unsupported output [71], [80]
- e.g. invented standard citation, incorrect material property, flawed equation
- Trace claims to original, authoritative sources
- A fluent response or citation label is not evidence

Pass outputs through a verification gate



- Check with known numerical cases, units and assumptions
- Gate outputs before a report, calculation or system action

“Bias” has three distinct meanings

Term	Meaning
Data-coverage bias	Relevant machines, people or conditions missing or underrepresented
Social fairness	Unequal performance or impact across people or groups
Neural-network bias term	An additive model parameter

> Use precise language [81]. Each needs different analysis; one is not evidence about another.

Protect information and connected systems

- Share restricted personal, assessment or proprietary data only with permission and safeguards
- Follow institutional disclosure and information-handling policies
- Generated code and connected tools can introduce security risks
- Review them before integrating; use source control and logging

“Human in the loop” needs specific responsibility

1. Name the reviewer and decision owner [9], [82]
2. State which outputs require review
3. Define what evidence must be checked
4. Give an authorized person power to pause or stop
5. Keep an audit trail: provenance, assumptions, units, baselines, decisions

Scale the method and disclose assistance

- AI computation consumes physical resources [83]
- Scale methods to the task and resource constraints
- Avoid unsupported numerical claims about a tool’s footprint
- Disclose AI assistance as institutional policy requires

Keep a verifiable reasoning trail

- Retain a trail a reviewer can check
- Mark the supplied evidence
- Mark the generated suggestions
- Mark the engineer’s final decisions

CHECK YOUR UNDERSTANDING

What evidence and disclosure would you attach to an AI-assisted lab submission?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What evidence and disclosure would you attach to an AI-assisted lab submission?

> State the AI assistance used, as policy requires. Trace claims to authoritative sources, show known-case, unit and assumption checks, and keep a reasoning trail separating evidence, suggestions and decisions.

CHECK YOUR UNDERSTANDING

Why is “a human in the loop” not enough on its own?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why is “a human in the loop” not enough on its own?

> Oversight works only when responsibility is specific: a named reviewer, defined checks and the authority to pause or stop an action.

Key takeaways

- Trace AI claims to authoritative sources; check equations with known cases
- Data-coverage bias, social fairness and the bias term differ
- Do not share restricted data without permission and safeguards
- Oversight needs a named reviewer, defined checks and stop authority
- Disclose AI assistance as required; keep a verifiable reasoning trail

References

[9] National Institute of Standards and Technology, “Artificial Intelligence Risk Management Framework (AI RMF 1.0),” NIST AI 100-1, Gaithersburg, MD, USA, 2023. doi: 10.6028/NIST.AI.100-1

[71] Z. Ji et al., “Survey of hallucination in natural language generation,” ACM Comput. Surv., vol. 55, no. 12, pp. 1–38, 2023. doi: 10.1145/3571730

[80] National Institute of Standards and Technology, “Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile,” NIST AI 600-1, Gaithersburg, MD, USA, Jul. 2024. doi: 10.6028/NIST.AI.600-1

[81] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, “A survey on bias and fairness in machine learning,” ACM Comput. Surv., vol. 54, no. 6, pp. 1–35, 2022. doi: 10.1145/3457607

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[83] E. Strubell, A. Ganesh, and A. McCallum, “Energy and policy considerations for deep learning in NLP,” in Proc. 57th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2019, pp. 3645–3650. doi: 10.18653/v1/P19-1355

Numbers match the course reference list. The written lesson is the authoritative text.